

CHEM 101 Topic 3: Matter and Energy — Answer Key

Section 1: States of Matter & Phase Changes

Problem 1: Gas

Problem 2: Sublimation; Energy is absorbed

Problem 3: Dehydration reduces fluid for sweat; less evaporation = less cooling

Section 2: Classifying Matter

Problem 4: Compound (H and O chemically combined)

Problem 5: Heterogeneous mixture

Problem 6: Distillation

Section 3: Physical vs. Chemical Changes

Problem 7: a) Physical, b) Chemical, c) Physical, d) Chemical

Problem 8: Chemical change (heat production)

Section 4: Conservation of Mass and Energy

Problem 9: 16.0 g

Problem 10: Energy is conserved; chemical energy transforms to heat and kinetic energy